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HISTORY · Advent of Europeans and the rise of British power

UNDERSTAND IT FIRST

What is it? From **1498** European trading companies came to India **by sea**, first to buy **spices and cotton cloth**. Over about **260 years** one of them, the **English East India Company**, defeated its European rivals and Indian rulers and turned from a **trader into a ruler**. The reasons are **why they came, how they competed, and why the British won**. Learn it as **Why Europe came → Portuguese → Dutch → English → Danes and French → Carnatic Wars → Bengal (Plassey, Buxar) → Why the British won**.

European companies in India

1. Why did Europe come by sea?

- **Spices and cloth** (pepper, cloves, cinnamon, Indian cotton) were very costly in Europe, and the old trade went through **Arab and Venetian middlemen**. The **Ottoman capture of Constantinople (1453)** made the land route harder and dearer.
- **Renaissance, science and new ships** (compass, astrolabe, the **caravel**, maps) and royal backing (**Prince Henry the Navigator** of Portugal) made long voyages possible. **Mercantilism** (the idea that a country grows rich by selling more than it buys, and by colonies) pushed governments to back trade companies.

2. The Portuguese

Person / event	Facts
Vasco da Gama (1498)	Reached **Calicut** (Kerala) on **20 May 1498** via the Cape of Good Hope, helped by a **Gujarati pilot**; welcomed by the **Zamorin**; returned with a cargo worth sixty times the cost of the voyage.
Francisco de Almeida (1505–09)	First Viceroy; **'Blue Water Policy'**: control the **sea** and not build many land forts; won the **Battle of Diu (1509)** against Egypt, Turkey and Gujarat's fleet.
Alfonso de Albuquerque (1509–15)	Real founder of Portuguese power; captured **Goa from Bijapur (1510)**, **Malacca (1511)**, **Hormuz (1515)**; encouraged **marriages** with Indian women.
Nino da Cunha (1529–38)	Moved the capital from **Cochin to Goa** (1530); took **Diu and Daman** and **Bassein (1534–35)**.
Methods and effects	**Cartaz** (naval pass for ships); **Jesuits** (St **Francis Xavier**, 1542); **Goa Inquisition** (1560); first **printing press** (Goa, 1556); new crops: **tobacco, maize, potato, chilli, cashew, pineapple, papaya**; **Santhome** (Mylapore) settlement. **Decline:** Dutch and English rivalry; **Marathas took Salsette and Bassein (1739)**; **Goa, Daman and Diu** freed by India in **1961**.

3. The Dutch

- **United East India Company (VOC)**, 1602. Main interest: the **Spice Islands (Indonesia)**; in India **cotton textiles**, indigo, saltpetre and opium. **Factories: Masulipatnam (1605)**, the **first Dutch factory in India**; **Pulicat (1610)** with **Fort Geldria**; **Surat (1616)**, **Chinsura (1653, Bengal)**, **Nagapattinam**; took **Cochin from the Portuguese (1663)**.
- **Decline:** English naval power; **Travancore's Marthanda Varma defeated them at Colachel (1741)**; the British beat them at **Bedara (Chinsura), 1759**. They focused on Indonesia and left India.

4. The English East India Company

Year	Event
1600	**Queen Elizabeth I** gave a charter (31 December) to the **'Governor and Company of Merchants of London Trading into the East Indies'**, with a **monopoly of eastern trade**.
1608–15	**Captain Hawkins (1608)** at Jahangir's court failed; **Thomas Best** beat the Portuguese at **Swally (1612)**; **Sir Thomas Roe (1615)** got **trading rights**; **Surat factory (1613)**.
1611–1625	**Masulipatnam (1611)**, the **first English factory in south-east India** (Andhra); **Pettapoli**; **Armagon (Durgarajapatnam), 1625**.
1639–40	**Francis Day** and **Andrew Cogan** got land at **Madras** from the local Nayaka (**Damarla Venkatapathi**) with the permission of the **Chandragiri ruler**; built **Fort St George**.
1651–1690	**Gabriel Boughton** treated a Mughal princess and won concessions in **Bengal (1651)**; **Bombay**, Charles II's dowry from **Catherine of Braganza (1661)**, leased to the Company in **1668**; **Job Charnock** founded **Calcutta (1690)**; **Fort William (1696–1700)**.
1717	Emperor **Farrukhsiyar's farman**: duty-free trade in Bengal for a payment of ₹3,000 a year; the Company's **dastak** (pass) system, often misused. Called the **'Magna Carta of the Company'**.

5. The Danes and the French

- **Danes:** company founded **1616**; **Tranquebar (Tharangambadi), 1620**; **Serampore (1755)**; both **sold to the British in 1845**; known more for **missionary work** (Serampore Mission).
- **French: Compagnie des Indes Orientales (1664)**, founded under **Colbert** (Louis XIV's minister) as a **State-run** company; **Surat (1668)**, **Masulipatnam (1669)**; **Pondicherry (1673)** founded by **Francois Martin**; **Chandernagore (1690)**; **Mahe**, **Karaikal** and **Yanam** (Andhra, 1723). **Dupleix** (governor 1742–54) used **Indian soldiers trained in European style (sepoys)** and **interfered in Indian successions**. French territories joined India in **1954** (de facto) and **1962** (de jure).

6. The Carnatic Wars (Anglo-French rivalry)

War	Facts

First (1746–48)	Extension of the **Austrian Succession War**; French took **Madras (1746)**; **Battle of St Thome (Adyar)**: a small French force beat the Nawab of the Carnatic **Anwaruddin**, showing how **disciplined European troops** could beat large Indian armies; **Treaty of Aix-la-Chapelle (1748)** returned Madras.
Second (1749–54)	Quarrels over succession in **Hyderabad** (Nasir Jung vs Muzaffar Jung) and the **Carnatic** (Anwaruddin vs Chanda Sahib); French (**Dupleix, Bussy**) backed Chanda Sahib and Muzaffar Jung; English backed **Muhammad Ali**; **Clive's siege of Arcot (1751)**; **Treaty of Pondicherry (1754)**; Dupleix recalled. The French got the **Northern Circars (1753)** from the Nizam through **Bussy**.
Third (1756–63)	Part of the **Seven Years' War**; **Battle of Wandiwash (22 January 1760)**: **Eyre Coote** defeated **Lally**; **Pondicherry fell (1761)**; **Treaty of Paris (1763)** returned French factories but only as **trading posts**; in the Northern Circars **Col. Forde** beat the French at **Condore (1758)** and took **Masulipatnam (1759)**.

Why the French lost: weaker **navy**; **State control** and little support from the French government; **Dupleix's recall**; the English held **Bengal's wealth** after 1757; poorer leadership against **Clive and Eyre Coote**; the **Seven Years' War** in Europe went against France.

7. Bengal: Plassey and Buxar

From trader to ruler

Event	Facts
Background	**Alivardi Khan** died (1756); his grandson **Siraj-ud-Daulah** became Nawab. He was angry that the English **fortified Calcutta** without permission, **misused dastaks** and sheltered a runaway (**Krishna Das**). He took **Fort William (June 1756)**; the **Black Hole** incident (20 June 1756; Holwell's account of 146 prisoners and 123 deaths is **disputed**).
Plassey (23 June 1757)	**Clive** and Admiral Watson retook Calcutta (January 1757), then plotted with **Mir Jafar** (the Nawab's commander), **Rai Durlabh**, banker **Jagat Seth** and **Omichand**. At Plassey, Mir Jafar's troops stood aside; the Nawab was beaten in **a few hours** and later killed; **Mir Jafar** became Nawab and paid huge sums; the Company got **24 Parganas**. **Importance:** a win through **intrigue** that gave the Company **control of Bengal's wealth** and the base to defeat the French.
Mir Qasim (1760–63)	Made Nawab for ceding **Burdwan, Midnapore and Chittagong**; moved to **Munger**, built a modern army and **abolished internal duties** (which hurt Company traders); clash with the Company.
Buxar (22 October 1764)	**Hector Munro** defeated the combined armies of **Mir Qasim**, **Shuja-ud-Daulah** of Awadh and the Mughal Emperor **Shah Alam II**. More **decisive than Plassey**: it was a **fair fight** won by arms, making the Company the **leading power in north India**.
Treaty of Allahabad (1765)	**Clive** got from **Shah Alam II** the **Diwani (right to collect revenue) of Bengal, Bihar and Orissa** for **₹26 lakh** a year to the emperor; Shuja paid ₹50 lakh; **Kora and Allahabad** went to the emperor. **Dual government (1765–72)**: the Company had **revenue** (diwani) and the Nawab had

	administration (nizamat) but **no power to run it**: 'power without responsibility' for the Company, ended by **Warren Hastings (1772)**.

8. Mysore, subsidiary alliance and the Andhra region

Topic	Facts
Mysore Wars	**Haidar Ali** and **Tipu Sultan** resisted. **1st (1767–69)**: Treaty of Madras; **2nd (1780–84)**: Treaty of Mangalore; **3rd (1790–92)**: Cornwallis; **Treaty of Seringapatam (1792)**: Tipu lost **half his territory** and paid **3 crore** indemnity; **4th (1799)**: Tipu died defending **Seringapatam**; Wodeyars restored under Wellesley.
Subsidiary Alliance (Wellesley, 1798)	The Indian ruler accepted a **British force** at his cost, **no other European** in his service, **no war or treaty without Company's consent**, and a **British Resident** at his court. First to accept: **Hyderabad (1798)**, then **Mysore (1799)**, **Tanjore**, **Awadh (1801)**, **Peshwa (Bassein 1802)**, **Bhonsle and Scindia (1803–04)**.
Andhra region	**Northern Circars** (coastal districts from Srikakulam to Guntur): the French held them (Bussy, 1753), the British took them in **1759–65** and **Clive** obtained them from **Shah Alam II (1765)** and from the **Nizam (1766)**; **Guntur Circar** in **1788**. **Ceded districts** (**Kadapa, Kurnool, Anantapur, Bellary**): given to the Company by the **Nizam in 1800** under the Subsidiary Alliance. **Hyderabad (Telangana)** stayed a **princely State** under the Nizam until **1948**.

9. Why the British won; impact

Area	Points
Politics	**Naval strength**; **disciplined, drilled army** with better guns; **Indian disunity** and quarrels over succession; skill in **diplomacy and intrigue**; capable leaders (Clive, Hastings, Wellesley); **Indian sepoys** formed most of the army.
Economy	**Joint-stock company** with strong funds and **private investors**; **Bengal's revenue** (diwani) paid for wars; **Industrial Revolution** later gave better arms; Indian **cotton exports** fell after 1800s and the **'drain of wealth'** began.
Society and religion	**Missionaries** (Portuguese Jesuits, Danish Serampore Mission) began conversions and **printing and education**; European **towns and forts** (Madras, Bombay, Calcutta) grew; **new crops** and new ideas arrived.

EXAM TRAP

1498 Da Gama; **1510** Goa (Albuquerque); **1600** English charter; **1602** Dutch VOC; **1605** Dutch Masulipatnam; **1611** English Masulipatnam; **1613** Surat; **1616** Danes; **1639** Madras; **1664** French; **1673** Pondicherry; **1717** farman; **Carnatic Wars 1746–63**, **Wandiwash 1760**; **Plassey 23 Jun 1757**; **Buxar 22 Oct 1764**; **Allahabad 1765** (Diwani); **1792** Seringapatam; **1798** Hyderabad first subsidiary alliance; **1799** Tipu dies; **Northern Circars 1765**; **Ceded districts 1800**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the European companies in India:

1. The Portuguese under Albuquerque captured Goa from the Bijapur Sultan in 1510.

2. The first English factory in south-east India was set up at Masulipatnam in 1611.

3. The Dutch set up their first Indian factory at Surat.

4. The French founded Pondicherry in 1673 under Francois Martin.

Which of the above are correct?

A. 1, 2 and 4 only

B. 1 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. The first Dutch factory in India was at Masulipatnam (1605), not Surat (1616).

PRACTICE

Match List-I (event) with List-II (year):

List-I: A. Battle of Plassey B. Battle of Buxar C. Battle of Wandiwash D. Treaty of Allahabad

List-II: 1. 1760 2. 1757 3. 1765 4. 1764

A. A-2, B-1, C-4, D-3

B. A-4, B-2, C-1, D-3

C. A-2, B-4, C-3, D-1

D. A-2, B-4, C-1, D-3

Answer: D. Plassey 1757; Buxar 1764; Wandiwash 1760; Allahabad 1765.

PRACTICE

Which one of the following statements is NOT correct?

A. The Treaty of Allahabad (1765) gave the Diwani of Bengal, Bihar and Orissa to the Company.

B. The Travancore king Marthanda Varma defeated the Dutch at Colachel in 1741.

C. The Battle of Plassey was fought between the Company and Mir Qasim.

D. The Treaty of Paris (1763) ended the Third Carnatic War.

Answer: C. Plassey (1757) was fought against Siraj-ud-Daulah; Mir Qasim fought at Buxar (1764).

PRACTICE

Assertion (A): The French failed to build an empire in India despite early successes under Dupleix.
Reason (R): The British had stronger naval power and, after Plassey, the resources of Bengal.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Sea power and Bengal's revenue gave the British the advantage.

PRACTICE

A European company set up its first Indian factory at Masulipatnam in 1605 and later built Fort Geldria at Pulicat. Which company was it?

A. The Dutch East India Company

B. The French East India Company

C. The Portuguese Estado da India

D. The Danish East India Company

Answer: A. The Dutch VOC (1602) had its first factory at Masulipatnam (1605) and Fort Geldria at Pulicat.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the arrival order and key facts of each European company. 2. Write the three Carnatic Wars and why the French lost. 3. Write Plassey, Buxar, the Treaty of Allahabad and the Andhra link (Circars, Ceded districts).

POLITY · Judiciary: Supreme Court, High Courts; judicial review and judicial activism

UNDERSTAND IT FIRST

What is it? The judiciary is the organ that **settles disputes**, **protects the rights of citizens** and **tests whether laws and government actions follow the Constitution**. India has a **single, integrated** system with the **Supreme Court at the top**. Its two powerful tools are **judicial review** (striking down laws that break the Constitution) and, in recent decades, **judicial activism** (the courts actively filling gaps to protect rights).

The integrated judiciary

1. The Supreme Court

Point	Details
Composition (Art 124)	**Chief Justice of India and not more than 33 other judges** (34 in all, raised by the **2019 Act**); seat at **Delhi** (Art 130); **ad hoc judges** (Art 127) and **retired judges** (Art 128) can be called.
Qualifications	Citizen; **5 years as a High Court judge**, or **10 years as a High Court advocate**, or a **distinguished jurist** in the President's opinion.
Appointment	By the **President** after consulting judges: the **collegium system** (the CJI and four senior-most judges). **First Judges case (S.P. Gupta, 1981)**: executive's view prevails; **Second (1993)**: **CJI's view (with collegium) is binding**; **Third (1998)**: collegium of **five**. The **99th Amendment and NJAC** (2014) were **struck down in 2015** (Fourth Judges case) as against the basic structure and judicial independence.
Tenure and removal	Retire at **65**. Removed by the **President** after an **address by each House** (**special majority**: majority of total membership and **2/3 of members present and voting**) for **proved misbehaviour or incapacity** (**Art 124(4)**; **Judges (Inquiry) Act, 1968**). No judge has been removed: **V. Ramaswami (1993)** motion failed in the Lok Sabha.
Independence	**Salaries charged** on the Consolidated Fund (not voted); **no discussion** of conduct in Parliament (Art 121); **contempt** power (Art 129); no practice after retirement before any court; **separation from the executive** (Art 50).

Jurisdiction of the Supreme Court

Type	Details
Original (Art 131)	**Exclusive**: disputes between **Centre and States**, or between **States**; (not for disputes under pre-Constitution treaties).
Writ (Art 32)	**Right to Constitutional Remedies**: **habeas corpus, mandamus, prohibition, certiorari, quo warranto** to enforce **Fundamental Rights**; Dr Ambedkar called it the **'heart and soul'** of the Constitution.
Appellate (Art 132–136)	**Constitutional**, **civil** and **criminal** appeals from High Courts; **Art 136: Special Leave Petition (SLP)** to appeal from **any court or tribunal** (except armed forces).

Advisory (Art 143)	The **President** can ask for its opinion on a question of law or fact; the Court **need not** answer and the opinion is **not binding**.
Review, curative (Art 137)	It can **review its own judgment**; the **curative petition** (**Rupa Ashok Hurra, 2002**) is the last remedy against a miscarriage of justice.
Other	**Court of record** (Art 129); law declared is **binding on all courts** (Art 141); **Art 142**: orders **for complete justice**; all authorities must **aid** the Court (Art 144).

2. The High Courts and subordinate courts

Point	Details
Establishment	**Art 214**: a High Court for each State; **25 High Courts** (a few serve more than one State/UT); **Andhra Pradesh High Court** sits at **Amaravati** (since January 2019) and the **Telangana High Court** at **Hyderabad** (separate from 1 January 2019). **Art 215**: High Courts are **courts of record**.
Appointment, tenure	Chief Justice appointed by the **President** after consulting the **CJI**, the **Governor** and the High Court's Chief Justice; **qualifications**: citizen, **10 years a judicial officer** or **10 years a High Court advocate**; **retire at 62**; removed in the **same way as Supreme Court judges**. Transfer: **Art 222**.
Writs (Art 226)	Issues the five writs for **Fundamental Rights and 'for any other purpose'**, so its writ power is **wider than the Supreme Court's Art 32**.
Supervision (Art 227, 235)	**Superintendence over all courts and tribunals** in its area (Art 227); **control over subordinate courts** (posting, promotion, discipline: Art 235).
Subordinate courts (Art 233–237)	**District Judge** appointed by the **Governor** in consultation with the **High Court**; district and sessions courts, civil and criminal courts below; also **Family Courts**, **Nyaya Panchayats**, **Gram Nyayalayas (2008)**, **Lok Adalats** (Legal Services Authorities Act, 1987; **Art 39A free legal aid**) and **Tribunals** (**Art 323A, 323B**; **L. Chandra Kumar, 1997**: tribunals' decisions can be tested in High Courts).

3. Judicial review

- **Meaning:** the power of the courts to **examine** whether a **law or executive action** is **consistent with the Constitution**, and to declare it **void** if it is not. **Basis:** the **written, supreme Constitution** and federalism; **Art 13** (laws against Fundamental Rights are void), **Arts 32 and 226**, **131–136**. (In the USA it began with **Marbury v Madison, 1803**.)
- **Landmark cases: A.K. Gopalan (1950)**: narrow reading of Art 21; **Golaknath (1967)**: Parliament cannot amend Fundamental Rights; **Kesavananda Bharati (1973)**: Parliament may amend but **not the basic structure**; **Indira Gandhi v Raj Narain (1975)**; **Minerva Mills (1980)**: **judicial review and the balance of Fundamental Rights and Directive Principles** are basic structure; **L. Chandra Kumar (1997)**: judicial review is basic structure; **I.R. Coelho (2007)**: laws in the **Ninth Schedule** after 24 April 1973 can be reviewed.

4. Judicial activism and PIL

- **Judicial activism:** the courts **go beyond** deciding disputes and take a **proactive role** to protect rights and fill gaps left by the legislature and executive. Its tool is the **Public Interest Litigation (PIL)**: any public-spirited person can approach the court for **those who cannot**, relaxing the rule of **locus standi** (who may sue); even a **letter or postcard** can be treated as a petition (**epistolary jurisdiction**). It began in the late 1970s under Justices **P.N. Bhagwati** and **V.R. Krishna Iyer**.
- **Key cases: Hussainara Khatoon (1979)**: rights of **undertrial prisoners**, speedy trial; **Maneka Gandhi (1978)**: Art 21's procedure must be **fair, just and reasonable**; **Olga Tellis (1985)**: **right to livelihood**; **M.C. Mehta**: **environment**, **polluter pays**; **Vishaka (1997)**: guidelines against **sexual harassment at work**; **NALSA (2014)**: recognised **transgender** persons; **Puttaswamy (2017)**: **right to privacy** is a Fundamental Right; **Shayara Bano (2017)**: set aside instant **triple talaq**.
- **Criticism: judicial overreach** (courts entering the domain of the legislature and executive, against the **separation of powers**), slow justice, a **huge backlog** of cases, and misuse of PIL for **'publicity interest'**. The remedy is **judicial restraint**: courts must respect their limits, and the **Constitution** is the final guide.

EXAM TRAP

Art 124: CJI + 33 = 34; **retire 65 (SC), 62 (HC)**; **Art 124(4)** removal; **collegium**, **NJAC struck down 2015**; **Art 131** original; **Art 32** (SC) vs **Art 226** (HC, wider); **Art 136** SLP; **Art 143** advisory; **Art 141** binding law; **Art 142** complete justice; **Art 233** District Judge; **Art 235** HC control; **Art 39A** free legal aid; **Kesavananda 1973** basic structure; **Minerva Mills**; **PIL**: Hussainara Khatoon 1979, Vishaka 1997, Puttaswamy 2017.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the judges of the Supreme Court of India:
1. They retire at the age of 65.
2. They can be removed only by an address of both Houses of Parliament supported by a special majority, on grounds of proved misbehaviour or incapacity.
3. They are appointed by the President after consultation under the collegium system.
4. Their salaries are subject to the annual vote of Parliament.
Which of the above are correct?

A. 1, 2 and 3 only

B. 1 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: A. Judges' salaries are charged on the Consolidated Fund and are not subject to a vote.

PRACTICE

Match List-I (case) with List-II (principle):
List-I: A. Kesavananda Bharati B. Maneka Gandhi C. Vishaka D. Puttaswamy
List-II: 1. Right to privacy as a Fundamental Right 2. Basic structure doctrine 3. Guidelines against sexual harassment at the workplace 4. Procedure under Article 21 must be fair, just and reasonable

A. A-2, B-4, C-3, D-1

B. A-2, B-3, C-4, D-1

C. A-4, B-2, C-3, D-1

D. A-2, B-4, C-1, D-3

Answer: A. Kesavananda: basic structure; Maneka Gandhi: fair procedure; Vishaka: workplace harassment; Puttaswamy: privacy.

PRACTICE

Which one of the following statements is NOT correct?

A. The Supreme Court has exclusive original jurisdiction in disputes between the Centre and a State.

B. The District Judge is appointed by the Governor in consultation with the High Court.

C. The writ jurisdiction of a High Court under Article 226 is narrower than that of the Supreme Court under Article 32.

D. The advisory opinion of the Supreme Court under Article 143 is not binding on the President.

Answer: C. Art 226 can be used for Fundamental Rights and for any other purpose, so it is wider than Art 32.

PRACTICE

Assertion (A): The writ jurisdiction of the High Courts is wider than that of the Supreme Court.
Reason (R): Article 226 allows High Courts to issue writs for Fundamental Rights and for any other purpose, whereas Article 32 covers only Fundamental Rights.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. R correctly explains why A is true.

PRACTICE

A public-spirited person writes a letter to the Supreme Court about bonded labourers working in a quarry, and the Court treats the letter as a writ petition. This is an example of

A. Advisory jurisdiction

B. Public Interest Litigation (epistolary jurisdiction)

C. A curative petition

D. A review petition

Answer: B. Treating a letter as a petition is epistolary jurisdiction within PIL.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the Supreme Court's composition, appointment, removal and five jurisdictions. 2. Write the High Court's Art 226, 227, 235 and the subordinate courts. 3. Write judicial review cases and the PIL cases with one line each.

MATHS · Simple and compound interest

UNDERSTAND IT FIRST

What is it? Interest is the **price of using someone else's money**. The money lent is the **principal (P)**, the yearly percentage charged is the **rate (R)**, the period is the **time (T)** and the total repaid is the **amount (A = P + Interest)**. In **simple interest** the interest is calculated **only on the original principal**, so it grows in a **straight line**. In **compound interest** the interest of each period is **added to the principal**, so you pay **interest on interest**, and it grows **faster and faster**, like a snowball.

Simple and compound interest

1. Simple interest

- **SI = P × R × T / 100**, and **A = P + SI**. Also **P = (SI × 100) / (R × T)**, **R = (SI × 100) / (P × T)** and **T = (SI × 100) / (P × R)**. **T must be in years**: 6 months = 1/2 year, 9 months = 3/4 year, 73 days = 73/365 = 0.2 year.
- **Example:** ₹8,000 at 7.5% for 4 years: SI = 8,000 × 7.5 × 4 / 100 = **₹2,400**; A = ₹10,400.

- **Doubling and trebling:** at SI a sum becomes **n times** in T years at a rate **R = 100(n − 1)/T**. If it **doubles in 8 years**, **R = 12.5%**, and it will become **4 times in 24 years** (SI = 3P needs 3 × 8 years).
- **Two parts of a sum** lent at different rates: if the **total interest** is known, use **allegation** (weighted rates). **Borrowing at one rate and lending at another:** gain = P × (difference of rates) × T / 100.

2. Compound interest

Compounding	Amount A	Note
Yearly	**$A = P\,(1 + R/100)^{T}$**	**CI = A − P**
Half-yearly	**$A = P\,(1 + R/200)^{2T}$**	Rate **halves**, periods **double**
Quarterly	**$A = P\,(1 + R/400)^{4T}$**	Rate ÷ 4, periods × 4
Different rates R_1, R_2, R_3 in successive years	**$A = P\,(1 + R_1/100)(1 + R_2/100)(1 + R_3/100)$**	
Fractional years, e.g. 2½ years	$A = P\,(1 + R/100)^2 \times (1 + (R/2)/100)$	Compound for the whole years, then simple interest for the part year

EXAMPLE

CI − SI: for **2 years**, the difference is **$P \times (R/100)^2$**; for **3 years**, **$P \times (R/100)^2 \times (3 + R/100)$**. **Example:** ₹5,000 at 8% for 2 years: difference = $5{,}000 \times 0.08^2$ = **₹32**. **CI example:** ₹10,000 at 10% for 2 years: A = 10,000 × 1.1 × 1.1 = ₹12,100, so **CI = ₹2,100** (while SI would be ₹2,000: the extra ₹100 is the **interest on the first year's interest**). **Half-yearly:** ₹16,000 at 20% for 1 year: rate 10% per half-year, 2 periods: A = 16,000 × 1.1 × 1.1 = **₹19,360**. **Rule of 72:** at compound interest an amount **doubles in about 72 ÷ R years**.

EXAM TRAP

SI = PRT/100; **A = P + SI**; **CI: $A = P(1 + R/100)^{T}$**; **half-yearly: R/2, 2T**; **CI − SI (2 yrs) = $P(R/100)^2$**; **(3 yrs) = $P(R/100)^2(3 + R/100)$**; **SI doubling: T = 100/R**; **CI doubling ≈ 72/R**; **CI is always more than SI for T > 1**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

What is the simple interest on ₹8,000 at 7.5% per annum for 4 years?

A. ₹2,000

B. ₹2,400

C. ₹2,700

D. ₹3,000

Answer: B. SI = 8,000 × 7.5 × 4 / 100 = ₹2,400.

PRACTICE

What is the compound interest on ₹10,000 at 10% per annum for 2 years, compounded annually?

A. ₹2,000

B. ₹2,100

C. ₹2,200

D. ₹1,900

Answer: B. A = 10,000 × 1.1^2 = 12,100; CI = ₹2,100.

PRACTICE

What is the difference between the compound interest and the simple interest on ₹5,000 at 8% per annum for 2 years?

A. ₹24

B. ₹32

C. ₹40

D. ₹64

Answer: B. P $(R/100)^2$ = 5,000 × 0.0064 = ₹32.

PRACTICE

A sum of money doubles itself in 8 years at simple interest. In how many years will it become four times itself at the same rate?

A. 16 years

B. 24 years

C. 32 years

D. 20 years

Answer: B. Doubling needs SI = P in 8 years; four times needs SI = 3P, which takes 3 × 8 = 24 years.

PRACTICE

What is the amount on ₹16,000 at 20% per annum for 1 year if the interest is compounded half-yearly?

A. ₹19,200

B. ₹19,360

C. ₹19,520

D. ₹19,000

Answer: B. 10% per half-year for 2 periods: 16,000 × 1.1 × 1.1 = ₹19,360.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write SI, A and the three rearranged forms. 2. Write the CI formulas for yearly, half-yearly and quarterly. 3. Write CI − SI for 2 and 3 years and solve one example each.

NOTEBOOK PAGE · Day 13: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Europeans and the rise of British power

Area	Write
Order of arrival	Portuguese 1498 → English 1600 → Dutch 1602 → Danes 1616 → French 1664
Portuguese	Da Gama 1498, Almeida (Blue Water), Albuquerque (Goa 1510), cartaz, Goa capital 1530, Jesuits, new crops, end 1961
English, Dutch, French	Charter 1600, Hawkins 1608, Roe 1615, Masulipatnam 1611, Madras 1639, Bombay 1668, Calcutta 1690, 1717 farman; Dutch Masulipatnam 1605, Pulicat; French Pondicherry 1673, Yanam
Carnatic Wars	1746–48, 1749–54, 1756–63; Dupleix, Bussy, Clive (Arcot 1751), Wandiwash 1760, Treaty of Paris 1763; why the French lost
Bengal	Black Hole 1756, Plassey 1757, Mir Jafar, Mir Qasim, Buxar 1764, Allahabad 1765, Diwani, dual government
Andhra link	Northern Circars 1765, Ceded districts 1800, Nizam's subsidiary alliance 1798

POLITY: Judiciary

- Supreme Court: Art 124 (34 judges), appointment (collegium, Three Judges cases, NJAC struck down 2015), removal Art 124(4), jurisdictions (Art 131, 32, 132–136, 143, 137), Art 141, 142. High Courts: Art 214, Art

226 vs 32, Art 227, Art 235.

- Judicial review (Art 13, Kesavananda 1973, Minerva Mills, L. Chandra Kumar, I.R. Coelho); judicial activism and PIL (Hussainara Khatoon, S.P. Gupta, Maneka Gandhi, Vishaka, Puttaswamy); limits: judicial restraint.

MATHS: Simple and compound interest

Write SI = PRT/100, A = P + SI; CI: A = P(1 + R/100)^T; half-yearly and quarterly forms; CI − SI for 2 and 3 years; doubling and trebling at SI; one worked example of each.